

Matthew Emmanuel T. Labrador

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EDUCATION

University of the Philippines Manila

Manila, PH

BS Computer Science, University Scholar | Expected May 2029

Current Cumulative GWA: 1.0375

PAREF Southridge School

Manila, PH

Senior High School, Class Valedictorian and Gold Medalist | May 2025

EXPERIENCE AND PROJECTS

Tumbang Preso (TUMP) | Gear Up NCR Esports Game Dev Challenge, 1st Place

2026

Godot 4, GDScript, Blender, ENet Authoritative-Host Multiplayer, Dedicated Linux Servers

- Won 1st Place; now representing NCR at the National Finals in General Santos City, backed by DOST and partner companies.
- Sole developer, empty project to shipped build in five days: 3D models, characters, map, UI, sound design, bots, physics, netcode, and the full codebase.
- Engineered authoritative-host multiplayer over ENet, resolving contact by host-side distance after measuring 16 of 36 engine overlap callbacks silently failing.
- Deployed dedicated online lobbies on a Linux VPS with UDP discovery and join codes, reaching players behind carrier-grade NAT.
- Owned marketing and pitch, delivering the presentation and Q&A that won the regional.

eGovMed | eGov Hackathon PH, 1 of 10 Winners (₱100,000)

2026

React, Vite, Node.js, Express, TypeScript, Redis, eGov API Suite, Vercel

- Selected as one of ten winning teams (₱100,000) with Bisaya Hackers for an AI triage system that assesses and routes patients before they queue.
- Developed full-stack across React and Node/Express, owning integration of eight national eGov APIs: triage, SSO identity, National ID eVerify, face liveness, messaging, payments, reporting, and chain anchoring.
- Gave every integration paired mock and live paths behind an environment switch, so it demos offline and refuses to boot in production while any service is mocked.
- Owned the pitch and presentation, delivering the live demo to the judging panel.

Diabetic Complication Swarm (GlycoSwarm AI) | AMD Developer Hackathon ACT II, Track 3

2026

Python, LangGraph, FastAPI, Next.js, Ollama, AMD MI300X GPU

- Lead developer of an international, cross-timezone team building a multi-agent diabetic complication screening system.
- Built it full-stack: a LangGraph StateGraph of four parallel agents each writing and running its own Python scoring code on NHANES lab data, a FastAPI service, and a Next.js front end.
- Served live inference with Gemma 4 26B on an AMD MI300X GPU via Ollama, with automatic failover to Fireworks GLM 5.2.
- Designed and delivered the live demo and pitch deck to the judges.

AI Application & LLM Development

2025–2026

Python, Gemma 3B, Qwen 3, LangChain, LangGraph, Anthropic API

- Deployed open LLMs locally (Gemma 3B, Qwen 3) to study end-to-end setup, inference, and multi-agent performance.
- Built automated summarizer applications on the Anthropic API using RAG and Agent-to-Agent (A2A) architectures.
- Ran structured prompt evaluations across model versions and documented the findings.

UPLB Code Wars

2025–2026

C/C++, Algorithms, Compiler-free Environment

- Solved complex algorithmic problems entirely in Notepad, with no compiler, internet, or AI assistance.

CHIP-8 Emulator

2026

C/C++, Raylib, System Architecture

- Engineered a robust CHIP-8 emulator from scratch, implementing core CPU instructions and memory management.
- Built a custom visual debugger using Raylib to track system state, registers, and memory for analytical debugging.

Heart Disease Prediction Model

2025

Python, pandas, Google Colab

- Trained a supervised learning model in Python to predict heart disease from patient features end-to-end.
- Cleaned data and performed feature engineering with pandas, handling missing values and categorical variables.

TECHNICAL SKILLS

Languages: Python, SQL, C, C++, TypeScript, GDScript, HTML

ML & Data: Feature Engineering, Evaluation Metrics (accuracy, precision, recall, F1), pandas, Prompt Engineering, Model Deployment

Agentic AI & Orchestration: LangChain, LangGraph, FastAPI, Multi-Agent Systems, Ollama

Methods & Tools: Statistical Analysis, Model Evaluation, Google Colab, Git, Linux/CLI, Raylib, Godot, Blender, Next.js, Vercel

Certifications: AMD Multiagent Systems Deployment, CS50 Intro to CS, AI Prompt Engineering, Python, C

Spoken Languages: Filipino (native), English, Latin

LEADERSHIP AND ACTIVITIES

Competitive Debater

2022–2025

Various National Tournaments, Philippines

Quarterfinalist at Ateneo Peace Debate; competed at UP Diliman Debates, XSDC, and ASDC.

Google Developer Group and UP Socomsci

2025–2026

Coordinated cross-functional teams on operations and event setup.

Math Team Captain

2022–2024

PAREF Southridge School, Muntinlupa, PH

Led tutoring sessions and mentored peers for local math competitions.

AWARDS

1st Place, Gear Up NCR Esports Game Dev Challenge (2026), NCR representative to Nationals | Winner, 1 of 10, eGov Hackathon PH (2026, ₱100,000) | DOST Undergraduate Scholar (2025–Present) | Top 5 Finalist, Olymphysics NCR (2025) | 5th Place, Philippine Statistics Quiz NCR (2025)